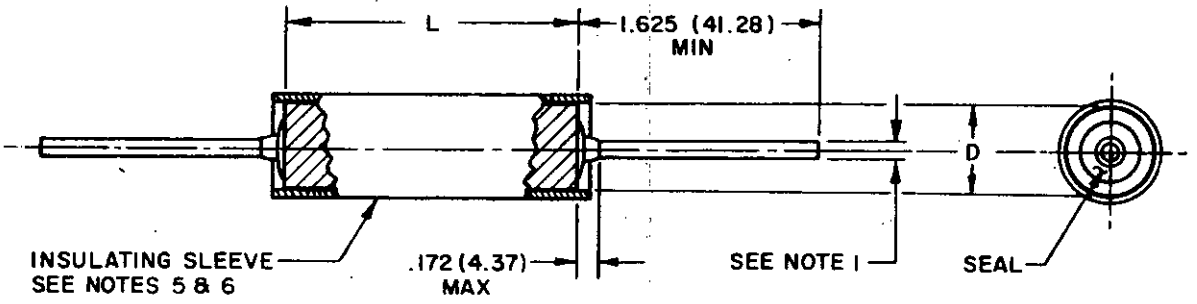
MILITARY SPECIFICATION SHEET

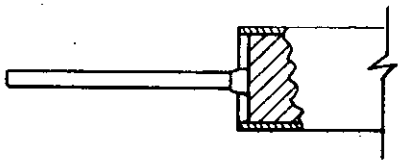
Ⓒ CAPACITORS, FIXED, METALLIZED, PAPER-PLASTIC FILM, OR PLASTIC FILM DIELECTRIC, DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CHR09 (INSULATED)

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39022.

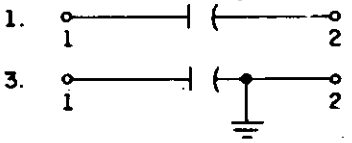
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



CASE FOR CIRCUIT 1



CASE FOR CIRCUIT 3



CIRCUIT DIAGRAM

NOTES:

1. Number 24 AWG wire for case diameters of .175 (4.45 mm) and .195 (4.95 mm). Number 22 AWG wire for case diameters of .235 (5.97 mm) and .312 (7.92 mm). Number 20 AWG wire for case diameters of .400 (10.16 mm) and over.
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents (to the nearest .01 mm) are given for general information only, and are based upon 1 inch = 25.4 mm.
5. Insulating sleeve shall extend beyond the capacitor body but shall not exceed .031 (.79 mm) inch on either end. Insulating sleeve thickness shall not exceed .016 (.41 mm) inch.
6. Plastic insulating sleeve shall be transparent; marking shall be applied to the capacitor case.
7. Metric equivalents are in parentheses.

Ⓒ denotes changes.

FSC 5910

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REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Case material: Nonmagnetic.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Rated voltage: See table I.

Dielectric material:
50 volt units normally polyethylene-terephthalate.
200, 400 and 600 volt units normally paper-polyethylene-terephthalate.

Operating temperature range:
50 volt units - -55° to +85°C.
200, 400 and 600 volt units - -55° to +125°C.

Failure rate level: L (5.0%), M (1.0%), P (0.1%), R (0.01%), and S (0.001%).

Radiographic inspection (X-ray): Method 209 of MIL-STD-202.

Temperature cycling: Method 102 of MIL-STD-202, condition D except step 3 shall be at applicable high test temperature.

Seal: Method 112 of MIL-STD-202, condition B, or (for liquid-filled units only) 1 hour at applicable high test temperature with no evidence of leakage.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202.
Terminal to terminal - 200 percent of rated voltage.
Terminals to case (when case is not a terminal) - 200 percent of rated voltage.

NOTE: For 100-percent inspection:
250 percent for not less than 5 seconds, or
200 percent for not less than 15 seconds.

Insulation resistance (IR): Method 302 of MIL-STD-202.
Terminal to terminal - See table III.
Terminals to case (when case is not a terminal) - 10,000 megohms, minimum.

Capacitance (Cap.): Method 305 of MIL-STD-202.

Dissipation factor (DF): 1 percent.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet).
125 percent of rated voltage applied: See MIL-C-39022 for voltage limitations.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G).
50 percent of rated voltage applied.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).
Salt solution - 5 percent.

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© TABLE I. Electrical characteristics, dimensions and dash numbers.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/ 2/		Dash number Failure rate level for (%/1,000 hr)				
volts	µF		Percent	L ±0.062	D ±0.031	L (5.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
50	0.018	1	5	0.750	0.175	1001	1261	1521	1781	2041
50	0.018	1	10	0.750	0.175	1002	1262	1522	1782	2042
50	0.018	3	5	0.688	0.175	1003	1263	1523	1783	2043
50	0.018	3	10	0.688	0.175	1004	1264	1524	1784	2044
50	0.022	1	5	0.750	0.175	1005	1265	1525	1785	2045
50	0.022	1	10	0.750	0.175	1006	1266	1526	1786	2046
50	0.022	3	5	0.688	0.175	1007	1267	1527	1787	2047
50	0.022	3	10	0.688	0.175	1008	1268	1528	1788	2048
50	0.027	1	5	0.750	0.175	1009	1269	1529	1789	2049
50	0.027	1	10	0.750	0.175	1010	1270	1530	1790	2050
50	0.027	3	5	0.688	0.175	1011	1271	1531	1791	2051
50	0.027	3	10	0.688	0.175	1012	1272	1532	1792	2052
50	0.033	1	5	0.750	0.175	1013	1273	1533	1793	2053
50	0.033	1	10	0.750	0.175	1014	1274	1534	1794	2054
50	0.033	3	5	0.688	0.175	1015	1275	1535	1795	2055
50	0.033	3	10	0.688	0.175	1016	1276	1536	1796	2056
50	0.039	1	5	0.750	0.195	1017	1277	1537	1797	2057
50	0.039	1	10	0.750	0.195	1018	1278	1538	1798	2058
50	0.039	3	5	0.688	0.195	1019	1279	1539	1799	2059
50	0.039	3	10	0.688	0.195	1020	1280	1540	1800	2060
50	0.047	1	5	0.750	0.195	1021	1281	1541	1801	2061
50	0.047	1	10	0.750	0.195	1022	1282	1542	1802	2062
50	0.047	3	5	0.688	0.195	1023	1283	1543	1803	2063
50	0.047	3	10	0.688	0.195	1024	1284	1544	1804	2064
50	0.056	1	5	0.750	0.235	1025	1285	1545	1805	2065
50	0.056	1	10	0.750	0.235	1026	1286	1546	1806	2066
50	0.056	3	5	0.688	0.235	1027	1287	1547	1807	2067
50	0.056	3	10	0.688	0.235	1028	1288	1548	1808	2068
50	0.068	1	5	0.750	0.235	1029	1289	1549	1809	2069
50	0.068	1	10	0.750	0.235	1030	1290	1550	1810	2070
50	0.068	3	5	0.688	0.235	1031	1291	1551	1811	2071
50	0.068	3	10	0.688	0.235	1032	1292	1552	1812	2072
50	0.082	1	5	0.750	0.235	1033	1293	1553	1813	2073
50	0.082	1	10	0.750	0.235	1034	1294	1554	1814	2074
50	0.082	3	5	0.688	0.235	1035	1295	1555	1815	2075
50	0.082	3	10	0.688	0.235	1036	1296	1556	1816	2076
50	0.10	1	5	0.750	0.235	1037	1297	1557	1817	2077
50	0.10	1	10	0.750	0.235	1038	1298	1558	1818	2078
50	0.10	3	5	0.688	0.235	1039	1299	1559	1819	2079
50	0.10	3	10	0.688	0.235	1040	1300	1560	1820	2080
50	0.12	1	5	0.750	0.235	1041	1301	1561	1821	2081
50	0.12	1	10	0.750	0.235	1042	1302	1562	1822	2082
50	0.12	3	5	0.688	0.235	1043	1303	1563	1823	2083
50	0.12	3	10	0.688	0.235	1044	1304	1564	1824	2084
50	0.15	1	5	0.875	0.312	1045	1305	1565	1825	2085
50	0.15	1	10	0.875	0.312	1046	1306	1566	1826	2086
50	0.15	3	5	0.812	0.312	1047	1307	1567	1827	2087
50	0.15	3	10	0.812	0.312	1048	1308	1568	1828	2088
50	0.18	1	5	0.875	0.312	1049	1309	1569	1829	2089
50	0.18	1	10	0.875	0.312	1050	1310	1570	1830	2090

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/ 2/		Dash number Failure rate level for (%/1, 000 hr)				
				L	D					
volts	μF		Percent	±0.062	±0.031	L (5.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
50	0.18	3	5	0.812	0.312	1051	1311	1571	1831	2091
50	0.18	3	10	0.812	0.312	1052	1312	1572	1832	2092
50	0.22	1	5	0.875	0.312	1053	1313	1573	1833	2093
50	0.22	1	10	0.875	0.312	1054	1314	1574	1834	2094
50	0.22	3	5	0.812	0.312	1055	1315	1575	1835	2095
50	0.22	3	10	0.812	0.312	1056	1316	1576	1836	2096
50	0.27	1	5	0.875	0.312	1057	1317	1577	1837	2097
50	0.27	1	10	0.875	0.312	1058	1318	1578	1838	2098
50	0.27	3	5	0.812	0.312	1059	1319	1579	1839	2099
50	0.27	3	10	0.812	0.312	1060	1320	1580	1840	2100
50	0.33	1	5	0.875	0.312	1061	1321	1581	1841	2101
50	0.33	1	10	0.875	0.312	1062	1322	1582	1842	2102
50	0.33	3	5	0.812	0.312	1063	1323	1583	1843	2103
50	0.33	3	10	0.812	0.312	1064	1324	1584	1844	2104
50	0.39	1	5	1.125	0.312	1065	1325	1585	1845	2105
50	0.39	1	10	1.125	0.312	1066	1326	1586	1846	2106
50	0.39	3	5	1.062	0.312	1067	1327	1587	1847	2107
50	0.39	3	10	1.062	0.312	1068	1328	1588	1848	2108
50	0.47	1	5	1.125	0.312	1069	1329	1589	1849	2109
50	0.47	1	10	1.125	0.312	1070	1330	1590	1850	2110
50	0.47	3	5	1.062	0.312	1071	1331	1591	1851	2111
50	0.47	3	10	1.062	0.312	1072	1332	1592	1852	2112
50	0.56	1	5	1.125	0.400	1073	1333	1593	1853	2113
50	0.56	1	10	1.125	0.400	1074	1334	1594	1854	2114
50	0.56	3	5	1.062	0.400	1075	1335	1595	1855	2115
50	0.56	3	10	1.062	0.400	1076	1336	1596	1856	2116
50	0.68	1	5	1.125	0.400	1077	1337	1597	1857	2117
50	0.68	1	10	1.125	0.400	1078	1338	1598	1858	2118
50	0.68	3	5	1.062	0.400	1079	1339	1599	1859	2119
50	0.68	3	10	1.062	0.400	1080	1340	1600	1860	2120
50	0.82	1	5	1.125	0.400	1081	1341	1601	1861	2121
50	0.82	1	10	1.125	0.400	1082	1342	1602	1862	2122
50	0.82	3	5	1.062	0.400	1083	1343	1603	1863	2123
50	0.82	3	10	1.062	0.400	1084	1344	1604	1864	2124
50	1.0	1	5	1.125	0.400	1085	1345	1605	1865	2125
50	1.0	1	10	1.125	0.400	1086	1346	1606	1866	2126
50	1.0	3	5	1.062	0.400	1087	1347	1607	1867	2127
50	1.0	3	10	1.062	0.400	1088	1348	1608	1868	2128
50	1.2	1	5	1.375	0.400	1089	1349	1609	1869	2129
50	1.2	1	10	1.375	0.400	1090	1350	1610	1870	2130
50	1.2	3	5	1.312	0.400	1091	1351	1611	1871	2131
50	1.2	3	10	1.312	0.400	1092	1352	1612	1872	2132
50	1.5	1	5	1.125	0.500	1093	1353	1613	1873	2133
50	1.5	1	10	1.125	0.500	1094	1354	1614	1874	2134
50	1.5	3	5	1.062	0.500	1095	1355	1615	1875	2135
50	1.5	3	10	1.062	0.500	1096	1356	1616	1876	2136
50	1.8	1	5	1.375	0.500	1097	1357	1617	1877	2137
50	1.8	1	10	1.375	0.500	1098	1358	1618	1878	2138
50	1.8	3	5	1.312	0.500	1099	1359	1619	1879	2139
50	1.8	3	10	1.312	0.500	1100	1360	1620	1880	2140

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/ 2/		Dash number Failure rate level for (%/1,000 hr)				
volts	μF		Percent	L ±0.062	D ±0.031	L (5.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
50	2.2	1	5	1.125	0.562	1101	1361	1621	1881	2141
50	2.2	1	10	1.125	0.562	1102	1362	1622	1882	2142
50	2.2	3	5	1.062	0.562	1103	1363	1623	1883	2143
50	2.2	3	10	1.062	0.562	1104	1364	1624	1884	2144
50	2.7	1	5	1.375	0.562	1105	1365	1625	1885	2145
50	2.7	1	10	1.375	0.562	1106	1366	1626	1886	2146
50	2.7	3	5	1.312	0.562	1107	1367	1627	1887	2147
50	2.7	3	10	1.312	0.562	1108	1368	1628	1888	2148
50	3.3	1	5	1.375	0.562	1109	1369	1629	1889	2149
50	3.3	1	10	1.375	0.562	1110	1370	1630	1890	2150
50	3.3	3	5	1.312	0.562	1111	1371	1631	1891	2151
50	3.3	3	10	1.312	0.562	1112	1372	1632	1892	2152
50	3.9	1	5	1.625	0.562	1113	1373	1633	1893	2153
50	3.9	1	10	1.625	0.562	1114	1374	1634	1894	2154
50	3.9	3	5	1.562	0.562	1115	1375	1635	1895	2155
50	3.9	3	10	1.562	0.562	1116	1376	1636	1896	2156
50	4.7	1	5	1.750	0.562	1117	1377	1637	1897	2157
50	4.7	1	10	1.750	0.562	1118	1378	1638	1898	2158
50	4.7	3	5	1.688	0.562	1119	1379	1639	1899	2159
50	4.7	3	10	1.688	0.562	1120	1380	1640	1900	2160
50	5.6	1	5	1.875	0.562	1121	1381	1641	1901	2161
50	5.6	1	10	1.875	0.562	1122	1382	1642	1902	2162
50	5.6	3	5	1.812	0.562	1123	1383	1643	1903	2163
50	5.6	3	10	1.812	0.562	1124	1384	1644	1904	2164
50	6.8	1	5	1.625	0.672	1125	1385	1645	1905	2165
50	6.8	1	10	1.625	0.672	1126	1386	1646	1906	2166
50	6.8	3	5	1.562	0.672	1127	1387	1647	1907	2167
50	6.8	3	10	1.562	0.672	1128	1388	1648	1908	2168
50	8.2	1	5	1.875	0.672	1129	1389	1649	1909	2169
50	8.2	1	10	1.875	0.672	1130	1390	1650	1910	2170
50	8.2	3	5	1.812	0.672	1131	1391	1651	1911	2171
50	8.2	3	10	1.812	0.672	1132	1392	1652	1912	2172
50	10.0	1	5	1.875	0.750	1133	1393	1653	1913	2173
50	10.0	1	10	1.875	0.750	1134	1394	1654	1914	2174
50	10.0	3	5	1.812	0.750	1135	1395	1655	1915	2175
50	10.0	3	10	1.812	0.750	1136	1396	1656	1916	2176
50	12.0	1	5	2.375	0.750	1137	1397	1657	1917	2177
50	12.0	1	10	2.375	0.750	1138	1398	1658	1918	2178
50	12.0	3	5	2.312	0.750	1139	1399	1659	1919	2179
50	12.0	3	10	2.312	0.750	1140	1400	1660	1920	2180
200	0.10	1	10	0.844	0.312	1141	1401	1661	1921	2181
200	0.10	1	20	0.844	0.312	1142	1402	1662	1922	2182
200	0.10	3	10	0.781	0.312	1143	1403	1663	1923	2183
200	0.10	3	20	0.781	0.312	1144	1404	1664	1924	2184
200	0.22	1	10	1.125	0.312	1145	1405	1665	1925	2185
200	0.22	1	20	1.125	0.312	1146	1406	1666	1926	2186
200	0.22	3	10	1.062	0.312	1147	1407	1667	1927	2187
200	0.22	3	20	1.062	0.312	1148	1408	1668	1928	2188
200	0.47	1	10	1.125	0.400	1149	1409	1669	1929	2189
200	0.47	1	20	1.125	0.400	1150	1410	1670	1930	2190

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/ 2/		Dash number Failure rate level for (%/1, 000 hr)				
volts	μF		Percent	L ±0.062	D ±0.031	L (5.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
200	0.47	3	10	1.062	0.400	1151	1411	1671	1931	2191
200	0.47	3	20	1.062	0.400	1152	1412	1672	1932	2192
200	1.0	1	10	1.125	0.562	1153	1413	1673	1933	2193
200	1.0	1	20	1.125	0.562	1154	1414	1674	1934	2194
200	1.0	3	10	1.062	0.562	1155	1415	1675	1935	2195
200	1.0	3	20	1.062	0.562	1156	1416	1676	1936	2196
200	1.5	1	10	1.844	0.562	1157	1417	1677	1937	2197
200	1.5	1	20	1.844	0.562	1158	1418	1678	1938	2198
200	1.5	3	10	1.781	0.562	1159	1419	1679	1939	2199
200	1.5	3	20	1.781	0.562	1160	1420	1680	1940	2200
200	2.2	1	10	1.844	0.562	1161	1421	1681	1941	2201
200	2.2	1	20	1.844	0.562	1162	1422	1682	1942	2202
200	2.2	3	10	1.781	0.562	1163	1423	1683	1943	2203
200	2.2	3	20	1.781	0.562	1164	1424	1684	1944	2204
200	3.3	1	10	1.875	0.670	1165	1425	1685	1945	2205
200	3.3	1	20	1.875	0.670	1166	1426	1686	1946	2206
200	3.3	3	10	1.812	0.670	1167	1427	1687	1947	2207
200	3.3	3	20	1.812	0.670	1168	1428	1688	1948	2208
200	4.7	1	10	1.875	1.000	1169	1429	1689	1949	2209
200	4.7	1	20	1.875	1.000	1170	1430	1690	1950	2210
200	4.7	3	10	1.812	1.000	1171	1431	1691	1951	2211
200	4.7	3	20	1.812	1.000	1172	1432	1692	1952	2212
200	6.8	1	10	1.875	1.000	1173	1433	1693	1953	2213
200	6.8	1	20	1.875	1.000	1174	1434	1694	1954	2214
200	6.8	3	10	1.812	1.000	1175	1435	1695	1955	2215
200	6.8	3	20	1.812	1.000	1176	1436	1696	1956	2216
200	10.0	1	10	2.375	1.000	1177	1437	1697	1957	2217
200	10.0	1	20	2.375	1.000	1178	1438	1698	1958	2218
200	10.0	3	10	2.312	1.000	1179	1439	1699	1959	2219
200	10.0	3	20	2.312	1.000	1180	1440	1700	1960	2220
200	12.0	1	10	2.625	1.000	1181	1441	1701	1961	2221
200	12.0	1	20	2.625	1.000	1182	1442	1702	1962	2222
200	12.0	3	10	2.562	1.000	1183	1443	1703	1963	2223
200	12.0	3	20	2.562	1.000	1184	1444	1704	1964	2224
400	0.047	1	10	1.125	0.312	1185	1445	1705	1965	2225
400	0.047	1	20	1.125	0.312	1186	1446	1706	1966	2226
400	0.047	3	10	1.062	0.312	1187	1447	1707	1967	2227
400	0.047	3	20	1.062	0.312	1188	1448	1708	1968	2228
400	0.10	1	10	1.125	0.400	1189	1449	1709	1969	2229
400	0.10	1	20	1.125	0.400	1190	1450	1710	1970	2230
400	0.10	3	10	1.062	0.400	1191	1451	1711	1971	2231
400	0.10	3	20	1.062	0.400	1192	1452	1712	1972	2232
400	0.22	1	10	1.125	0.562	1193	1453	1713	1973	2233
400	0.22	1	20	1.125	0.562	1194	1454	1714	1974	2234
400	0.22	3	10	1.062	0.562	1195	1455	1715	1975	2235
400	0.22	3	20	1.062	0.562	1196	1456	1716	1976	2236
400	0.47	1	10	1.625	0.562	1197	1457	1717	1977	2237
400	0.47	1	20	1.625	0.562	1198	1458	1718	1978	2238
400	0.47	3	10	1.562	0.562	1199	1459	1719	1979	2239
400	0.47	3	20	1.562	0.562	1200	1460	1720	1980	2240

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/ 2/		Dash number Failure rate level for (%/1, 000 hr)				
				L	D					
volts	μF		Percent	±0.062	±0.031	L (5.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
400	1.0	1	10	1.875	0.750	1201	1461	1721	1981	2241
400	1.0	1	20	1.875	0.750	1202	1462	1722	1982	2242
400	1.0	3	10	1.812	0.750	1203	1463	1723	1983	2243
400	1.0	3	20	1.812	0.750	1204	1464	1724	1984	2244
400	2.2	1	10	1.875	1.000	1205	1465	1725	1985	2245
400	2.2	1	20	1.875	1.000	1206	1466	1726	1986	2246
400	2.2	3	10	1.812	1.000	1207	1467	1727	1987	2247
400	2.2	3	20	1.812	1.000	1208	1468	1728	1988	2248
400	3.3	1	10	2.625	1.000	1209	1469	1729	1989	2249
400	3.3	1	20	2.625	1.000	1210	1470	1730	1990	2250
400	3.3	3	10	2.562	1.000	1211	1471	1731	1991	2251
400	3.3	3	20	2.562	1.000	1212	1472	1732	1992	2252
600	0.01	1	10	0.812	0.312	1213	1473	1733	1993	2253
600	0.01	1	20	0.812	0.312	1214	1474	1734	1994	2254
600	0.01	3	10	0.750	0.312	1215	1475	1735	1995	2255
600	0.01	3	20	0.750	0.312	1216	1476	1736	1996	2256
600	0.022	1	10	0.812	0.312	1217	1477	1737	1997	2257
600	0.022	1	20	0.812	0.312	1218	1478	1738	1998	2258
600	0.022	3	10	0.750	0.312	1219	1479	1739	1999	2259
600	0.022	3	20	0.750	0.312	1220	1480	1740	2000	2260
600	0.047	1	10	1.125	0.400	1221	1481	1741	2001	2261
600	0.047	1	20	1.125	0.400	1222	1482	1742	2002	2262
600	0.047	3	10	1.062	0.400	1223	1483	1743	2003	2263
600	0.047	3	20	1.062	0.400	1224	1484	1744	2004	2264
600	0.10	1	10	1.125	0.500	1225	1485	1745	2005	2265
600	0.10	1	20	1.125	0.500	1226	1486	1746	2006	2266
600	0.10	3	10	1.062	0.500	1227	1487	1747	2007	2267
600	0.10	3	20	1.062	0.500	1228	1488	1748	2008	2268
600	0.22	1	10	1.375	0.562	1229	1489	1749	2009	2269
600	0.22	1	20	1.375	0.562	1230	1490	1750	2010	2270
600	0.22	3	10	1.312	0.562	1231	1491	1751	2011	2271
600	0.22	3	20	1.312	0.562	1232	1492	1752	2012	2272
600	0.47	1	10	1.625	0.670	1233	1493	1753	2013	2273
600	0.47	1	20	1.625	0.670	1234	1494	1754	2014	2274
600	0.47	3	10	1.562	0.670	1235	1495	1755	2015	2275
600	0.47	3	20	1.562	0.670	1236	1496	1756	2016	2276
600	1.0	1	10	1.844	1.000	1237	1497	1757	2017	2277
600	1.0	1	20	1.844	1.000	1238	1498	1758	2018	2278
600	1.0	3	10	1.781	1.000	1239	1499	1759	2019	2279
600	1.0	3	20	1.781	1.000	1240	1500	1760	2020	2280
600	1.5	1	10	1.875	1.000	1241	1501	1761	2021	2281
600	1.5	1	20	1.875	1.000	1242	1502	1762	2022	2282
600	1.5	3	10	1.812	1.000	1243	1503	1763	2023	2283
600	1.5	3	20	1.812	1.000	1244	1504	1764	2024	2284
600	2.2	1	10	2.625	1.000	1245	1505	1765	2025	2285
600	2.2	1	20	2.625	1.000	1246	1506	1766	2026	2286
600	2.2	3	10	2.562	1.000	1247	1507	1767	2027	2287
600	2.2	3	20	2.562	1.000	1248	1508	1768	2028	2288

1/ Dimensions shown are bare case sizes (see figure).

2/ Metric equivalents see table II.

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TABLE II. Metric equivalents of decimal inches. 1/

Inches	MM	Inches	MM	Inches	MM
.031	.79	.750	19.05	1.688	42.88
.062	1.57	.781	19.84	1.750	44.45
.175	4.45	.812	20.62	1.781	45.24
.195	4.95	.844	21.44	1.812	46.02
.235	5.97	.875	22.23	1.844	46.84
.312	7.92	1.000	25.40	1.875	47.63
.400	10.16	1.062	26.97	2.312	58.72
.500	12.70	1.125	28.58	2.375	60.33
.562	14.27	1.312	33.32	2.562	65.07
.670	17.02	1.375	34.93	2.625	66.68
.672	17.07	1.562	39.67		
.688	17.48	1.625	41.28		

1/ Metric equivalents (to the nearest .01 millimeter) are given for general information only and are based upon 1 inch = 25.4 mm.

TABLE III. Characteristics.

Dielectric material	Paper-polyethylene-terephthalate		Polyethylene-terephthalate
Operating temperature range	-55 to +125°C		-55 to +85°C
DC voltage rating (volts)	200 V	400 V 600 V	50
High ambient test temperature Low ambient test temperature	125°+4, -0°C -55°+0, -3°C	125°+4, -0°C -55°+0, -3°C	85°+4, -0°C -55°+0, -3°C
Megohm x microfarads (minimum) At 25°C At high ambient test temperature	2,000 10	2,000 40	25,000 3,000
Insulation resistance (megohms) (need not exceed) At 25°C At high ambient test temperature	12,000 150	12,000 600	50,000 6,000
Capacitance change with temperature (maximum) At -55°C At high ambient test temperature	-10 +20	-10 +20	-8 percent +8 percent

Immersion: Method 104 of MIL-STD-202, condition C.
DWV:
Insulating sleeves - Greater than 4,000 volts. dc.
Terminal to terminal - 200 percent of rated voltage.
Terminals to case (when case is not a terminal) - 200 percent of rated voltage.
IR:
Insulating sleeves - 100 megohms. minimum.
Terminal to terminal - Not less than 50 percent of value specified in table III.
Terminals to case (when case is not a terminal) - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial value.
DF - Not more than 1.1 percent.

Solderability: Method 208 of MIL-STD-202.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I.
50 percent of rated voltage applied.

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Moisture resistance: Method 106 of MIL-STD-202.
DWV, IR, Cap., and DF - Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and D.
Condition A - Applied force 5 pounds.
Condition D - 3 rotations of 360 degrees.

Low temperature and cap. change with temperature:
Low temperature: -55°+0, -3°C for 48 ±4 hours with rated voltage applied.
Cap. change with temperature:
-55°+0, -3°C - See table III.
At applicable high test temperature - See table III.

Fault count: In accordance with MIL-C-39022.
At applicable high test temperature monitored for momentary breakdowns and periods of low IR.
Qualification - 2,000 +72, -0 hours; quality conformance (group C) 240+48, -0 hours.
Breakdowns - Shall not exceed 6 C or 1, whichever is greater (where C is the nominal capacitance in μF of the sample). Total counts for any single capacitor in a 24-hour period shall not exceed 4 C or 1, whichever is greater.
Period of low IR - Not to exceed 2 minutes.
IR:
Terminal to terminal - Not less than 33.3 percent of initial requirement.
Terminals to case (when case is not a terminal) - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial measured value.

Fungus: Method 508 of MIL-STD-810, procedure I.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B (260 ±5°C for 10 ±1 seconds).
IR - See table III.
Cap. - Within ±5 percent of initial value.
DF - Not more than 1.0 percent.

Life: Method 108 of MIL-STD-202.
Qualification:
Accelerated conditions: 140 percent of rated voltage for 2,000 +72, -0 hours.
Rated conditions: 100 percent of rated voltage for 10,000 +96, -0 hours.
DF (at applicable high test temperature) between 24 and 48 hours of test - 2 percent, maximum. During last 48 hours of test - 2.5 percent, maximum.
IR:
Insulating sleeves - 100 megohms, minimum.
Terminal to terminal - Not less than 33.3 percent of initial requirement.
Terminals to case (when case is not a terminal) - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial value.
DF (at +25° ±3°C) after life - 1.25 percent.

Quality conformance: 240 hours at accelerated conditions and meet the requirements for qualification test.

Extended life:
Accelerated conditions: 2,000 +72, -0 hours.
Rated conditions: 10,000 +96, -0 hours.

Superseded part numbers: The capacitors described herein are considered to be unilaterally interchangeable with those capacitors of the previous issue of this specification sheet.

Marking: In accordance with MIL-C-39022.

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Part number: M39022/01- (dash number from table I).

(C) Cross reference data from type designations of MIL-C-39022/1B to part numbers in MIL-C-39022/1C are not furnished since a qualified source did not exist at the time of this revision.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS, SH, CG
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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